QUBO Formulation: Minimum Multicut on a Tree

Minimum Multicut on a Tree

Problem Statement. Let $T = (V, E)$ be a tree with edge set E and non-negative edge weights w_e for each $e \in E$. Let $\mathcal{P} = \{(s_1, t_1), \ldots, (s_k, t_k)\}$ be a set of k terminal pairs, where $s_i, t_i \in V$. The Minimum Multicut on a Tree problem seeks a subset of edges $C \subseteq E$ such that for every pair (s_i, t_i), the nodes s_i and t_i are disconnected in the graph $T \setminus C$, and the total weight $\sum_{e \in C} w_e$ is minimized.

Decision Variables. For each edge $e \in E$, we introduce a binary decision variable $x_e \in \{0, 1\}$. The variable x_e indicates whether edge e is selected for the cut: $x_e = 1$ if $e \in C$, and $x_e = 0$ otherwise.

QUBO Derivation. Step 1: Objective Function. The objective is to minimize the total weight of the cut:

$$\min \sum_{e \in E} w_e x_e. \tag{1}$$

Step 2: Constraints. For each terminal pair (s_i, t_i), let $P_i \subseteq E$ denote the unique set of edges on the simple path between s_i and t_i in the tree. The separation constraint requires that at least one edge on this path is cut:

$$\sum_{e \in P_i} x_e \geq 1, \quad \forall i \in \{1, \ldots, k\}. \tag{2}$$

Step 3: Penalty Terms. We convert each inequality constraint to a quadratic penalty term. For a constraint $\sum_{e \in P_i} x_e \geq 1$, the violation occurs when $\sum_{e \in P_i} x_e = 0$. We use the penalty function $\lambda \left(1 - \sum_{e \in P_i} x_e\right)^2$, where $\lambda > 0$ is a penalty weight. Expanding this

expression and using the idempotent property $x_e^2 = x_e$ for binary variables:

$$\lambda \left(1 - \sum_{e \in P_i} x_e\right)^2 = \lambda \left(1 - 2\sum_{e \in P_i} x_e + \left(\sum_{e \in P_i} x_e\right)^2\right) \tag{3}$$

$$= \lambda \left(1 - 2\sum_{e \in P_i} x_e + \sum_{e \in P_i} x_e^2 + \sum_{e,f \in P_i, e \neq f} x_e x_f\right) \tag{4}$$

$$= \lambda \left(1 - 2\sum_{e \in P_i} x_e + \sum_{e \in P_i} x_e + \sum_{e,f \in P_i, e \neq f} x_e x_f\right) \tag{5}$$

$$= \lambda \left(1 - \sum_{e \in P_i} x_e + \sum_{e,f \in P_i, e \neq f} x_e x_f\right). \tag{6}$$

The term $\sum_{e,f \in P_i, e \neq f} x_e x_f$ contains each pair $\{e, f\}$ twice (once as $x_e x_f$ and once as $x_f x_e$). Thus, the quadratic contribution for distinct $e, f \in P_i$ is $2\lambda x_e x_f$. The linear contribution for $e \in P_i$ is $-\lambda x_e$, and the constant contribution is λ.

Step 4: QUBO Hamiltonian. Combining the objective and all penalty terms, the QUBO Hamiltonian $H(x)$ is:

$$H(x) = \sum_{e \in E} w_e x_e + \sum_{i=1}^{k} \lambda \left(1 - \sum_{e \in P_i} x_e + \sum_{e,f \in P_i, e \neq f} x_e x_f\right). \tag{7}$$

Step 5: Q Matrix and Offset. Collecting coefficients, the QUBO parameters are defined as follows. Let $n_e = |\{i : e \in P_i\}|$ be the number of terminal paths that include edge e, and let $n_{e,f} = |\{i : e, f \in P_i\}|$ be the number of terminal paths that include both edges e and f. The diagonal entries, off-diagonal entries, and offset are:

$$Q_{e,e} = w_e - \lambda n_e, \quad \forall e \in E, \tag{8}$$

$$Q_{e,f} = 2\lambda n_{e,f}, \quad \forall e, f \in E, e \neq f, \tag{9}$$

$$c = k\lambda. \tag{10}$$

The QUBO is then $H(x) = \sum_{e \in E} Q_{e,e} x_e + \sum_{e,f \in E, e<f} Q_{e,f} x_e x_f + c$.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any infeasible solution has a higher energy than the optimal feasible solution. A sufficient condition is $\lambda > \max_{e \in E} w_e$. This ensures that the penalty cost λ incurred by violating a separation constraint (which occurs when no edge on a path is cut) exceeds the maximum possible objective saving from omitting any single edge from the cut. Consequently, the optimizer is driven to satisfy all constraints.

Correctness Argument. Minimizing $H(x)$ over $x \in \{0,1\}^{|E|}$ recovers the optimal multi-cut. For any feasible solution, all constraints are satisfied, so the penalty terms vanish and

$H(x)$ equals the objective value. For any infeasible solution, at least one constraint is violated, incurring a penalty of at least λ. Since $\lambda > \max_{e \in E} w_e$, the penalty strictly dominates any potential reduction in the objective function, guaranteeing that the global minimum of $H(x)$ corresponds to a feasible solution with minimum objective value.

Illustrative Example. Consider a tree with two edges $E = \{e_0, e_1\}$ and weights $w_0 = 3$, $w_1 = 5$. Let there be a single terminal pair (s_1, t_1) such that the path between them includes both edges, i.e., $P_1 = \{e_0, e_1\}$. Here $k = 1$, $n_{e_0} = 1$, $n_{e_1} = 1$, and $n_{e_0,e_1} = 1$. Choosing $\lambda = 10$ (which satisfies $\lambda > \max\{3, 5\}$), the QUBO parameters are:

$$Q_{e_0,e_0} = 3 - 10(1) = -7, \tag{11}$$
$$Q_{e_1,e_1} = 5 - 10(1) = -5, \tag{12}$$
$$Q_{e_0,e_1} = 2(10)(1) = 20, \tag{13}$$
$$c = 1 \cdot 10 = 10. \tag{14}$$

The Hamiltonian is:

$$H(x) = -7x_0 - 5x_1 + 20x_0x_1 + 10. \tag{15}$$

Evaluating $H(x)$ for all $x \in \{0, 1\}^2$:

- $x = (0, 0)$: $H = 10$ (infeasible, penalty 10).
- $x = (1, 0)$: $H = -7 + 10 = 3$ (feasible, cost 3).
- $x = (0, 1)$: $H = -5 + 10 = 5$ (feasible, cost 5).
- $x = (1, 1)$: $H = -7 - 5 + 20 + 10 = 18$ (feasible, cost 8).

The minimum energy is 3 at $x = (1, 0)$, corresponding to cutting edge e_0 with total cost 3.